Experiment 2: Understanding image basic "image resize", image type conversion, extraction of colour bands, creating a synthetic image, ~~pseudo~~ pseudo colour image.

Image Resizing:

```
A = imread('cameraman.tif');
figure()
imshow(A), title('Original Image A: 256*256');
B = imresize(A, 0.5);
figure()
imshow(B), title('Output Image B: 128*128');
C = imresize(A, [128 512]);
figure()
imshow(C), title('Output Image C: 128*512');
```

Pseudo Image Creation

```
A = imread('onion.png');

figure()
imshow(A), title('Original Colour Image');

A_gray = rgb2gray(A);

figure()
imshow(A_gray), title('Gray Scale Image');

A_binary = A_gray>120;
figure()
imshow(A_binary), title('Binary Image');
A_ind = gray2ind(A_gray);
figure()
imshow(A_ind), title('Indexed Image');

map_64 = hsv(64);
A_pseudo_64 = ind2rgb(A_ind, map_64);

figure()
imshow(A_pseudo64), title('Pseudo Image
                                   hsv 64');
```

Psuedo Image Creation

```
A = imread('onion.png');
figure()
imshow(A), title('Original Colour Image');
A_gray = rgb2gray(A);
figure()
imshow(A_gray), title('Gray Scale Image');
A_binary = A_gray>120;
figure()
imshow(A_binary), title('Binary Image');
A_ind = gray2ind(A_gray);
figure()
imshow(A_ind), title('Indices Image');
map_64 = hsv(64);
A_pseudo_64 = ind2rgb(A_ind, map_64);
figure()
imshow(A_pseudo_64), title('Psuedo Image
                                  hsv 64');
```

Rotate an Image:

A = ~~imrotate~~ imread('onion.png');

figure()
imshow(A), title('Original Colour Image');

B = imrotate(A,45);

figure()
imshow(B), title('Output Image B');

Cropping an Image

A = imread('cameraman.tif');

figure()
imshow(A), title('Original Image A');

B = imcrop(A,[1 1 128 128]);

figure()
imshow(B), title('Output Image B');

Extraction of Colourmap and Pseudo Image Creation from RGB Image:

```
A = imread('onion.png');
figure()
imshow(A), title('Original Image A');
[A_ind, A_map] = rgb2ind(A, 16, 'nodither');
figure()
imshow(A_ind), title('Indices of Image A');

figure()
imshow(A_map), title('Colourmap of Image A');
map_64 = hsv(64);
A_pseudo_64 = ind2rgb(A_ind, map_64);
figure()
imshow(A_pseudo_64), title('Pseudo Image hsv 64');
```